Unit 5 – Polynomial Functions – Day 12 *Accelerated II/III*

Using the Fundamental Theorem of Algebra

Fundamental Theorem of Algebra: If $f(x)$ is a polynomial of degree n, where $n > 0$, then the equation $f(x) = 0$ has at least one root in the set of complex numbers.

**What this means:

Example 1: Find all the zeros of $f(x) = x^4 + 3x^3 - 8x^2 - 22x - 24$

Example 2: For the equation $f(x) = x^4 - 3x^3 + 4x + 1$, find the number of complex roots, the possible number of real roots, and the possible rational roots.

of complex roots =

of possible real roots =

Possible rational roots :

Example 3: For the equation $f(x) = 2x^5 + 10x^4 - 2x^2 - x + 3$, find the number of complex roots, the possible number of real roots, and the possible rational roots.

of complex roots =

of possible real roots =

Possible rational roots :